TANISH SANGHVI

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LinkedIn | GitHub | Portfolio

Professional Summary

Final-year Automation and Robotics Engineering student with hands-on experience building and deploying full-stack web applications, REST APIs, database-backed systems and automated data pipelines. Skilled in Python, JavaScript, TypeScript, React, Next.js, Flask and Node.js, with practical experience in authentication, role-based access control, automated testing and AI integration. National hackathon finalist who also led a four-member team on a final-year project.

Education

Vivekanand Education Society's Institute of Technology, Mumbai 2023 – 2027 (Expected)
B.E. in Automation and Robotics Engineering CGPA: 7.43/10.0
Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Machine Learning, Generative AI

Technical Skills

Languages: Python, JavaScript (ES6+), TypeScript, C++, SQL
Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS
Backend and APIs: Node.js, Express.js, Flask, REST API Design, Pydantic Validation
Databases: PostgreSQL, MongoDB, Firebase Firestore, Supabase
AI and Data: Gemini API Integration, NumPy, Data Pipelines, Data Visualisation
Testing and QA: Pytest, Jest, Playwright, Unit, Integration and End-to-End Testing, Debugging
Security: JWT Authentication, Role-Based Access Control, API Validation, Error Handling
Tools and Deployment: Git, GitHub, GitHub Actions (CI), Postman, Vercel, Render

Projects

Job Hunter – Automated Job Discovery Platform Live Demo | GitHub
Python, Flask, PostgreSQL, Supabase, GitHub Actions
- Built a modular job-ingestion pipeline with configurable parsing adapters for 114 company career-page sources across 9 applicant tracking system platforms.
- Integrated PostgreSQL/Supabase data persistence, configurable job filters and AI-assisted profile scoring to support job discovery and evaluation.
- Implemented error handling, exponential retry backoff and automated tests covering data extraction, persistence integrity and network-failure recovery.

Department Ledger Portal – Academic Records System Live Demo | GitHub
Next.js, React, Firebase Firestore, Jest, Playwright
- Developed an academic records platform with separate student, faculty and administrator workflows for record compilation and portfolio management.
- Implemented role-based access control, audit logging and document autofill using parsed data for structured academic record management.
- Wrote 199 Jest unit tests and 27 Playwright end-to-end tests to validate application logic and user workflows.

Edvanta – AI-Assisted Learning Platform Live Demo | GitHub
Python, Flask, React, MongoDB, Firebase, Pydantic
- Designed and deployed 33 modular REST endpoints supporting assessment workflows, AI-assisted assessment generation and personalized learning pathways.
- Integrated Gemini API to power AI-generated assessments and personalized learning experiences.
- Implemented Pydantic request validation, Firebase JWT authentication, protected-route middleware and role-based authorization for controlled API access.

DineEase – Restaurant Ordering System Live Demo | GitHub
React, Node.js, Express.js, MongoDB
- Developed a full-stack restaurant ordering application with menu browsing, cart management, checkout and administrator dashboards.
- Implemented role-protected administrator workflows and MongoDB-backed order processing with persistent application data.
- Wrote 35 automated tests covering order calculations, order-payload handling and route protection.

Battery State Estimator – Final-Year Major Project Live Demo | GitHub
Python, Flask, MongoDB, NumPy, Data Processing, Visualisation
- Developed a software-based battery simulation and telemetry pipeline generating voltage, current and temperature data across configurable drive cycles and fault scenarios.
- Implemented Echo State Network-based SOC/SOH estimation and comparative analysis against Extended Kalman Filter, Coulomb Counting and adaptive RLS methods.
- Integrated simulation outputs and results from 4 estimation methods with MongoDB and a Flask-based visualisation dashboard for monitoring battery behaviour and comparing model performance.

Achievements & Leadership

- **National Finalist – Hack Celestial 2.0:** Reached the national finals with Edvanta in a competition involving 320+ participating teams.
- **Co-organizer – UniMerge 1.0:** Coordinated technical operations, submission validation and developer support for a national hackathon with 130+ participants.
- **Team Lead – B.E. Major Project:** Led a 4-member team building a battery analytics platform, defining the system architecture, dividing modules across members and integrating simulation, estimation and visualisation components into one deployed system.